

May 13, 2026

1. (a) Let $A = (a_{ij})$ be an $n \times n$ matrix with entries $a_{ij} \in \{0, 1\}$. Prove the existence of a sign vector $x \in \{-1, 1\}^n$ such that $\|Ax\|_\infty = O(\sqrt{n \log n})$.
- (b) Show that there exists an $n \times n$ matrix B with 0/1 entries with the property $\|By\|_\infty \geq c\sqrt{n}$ for all $y \in \{-1, 1\}^n$, where $c > 0$ is a universal constant.

Hint: You may use Problem 4 in HW 2.

2. Let $0 < p \leq 1/2$ and let X be the random variable counting triangles in a randomly chosen graph G from $\mathcal{G}(n, p)$.

- (a) Show that $\mathbb{P}(X = 0) \geq (1 - p)^{\binom{n}{2}} \geq e^{-pn^2}$.

Hint: If a graph has no edges, then it's triangle-free.

- (b) Show that $\mathbb{P}(X = 0) \geq (1 - p^3)^{\binom{n}{3}} \geq e^{-p^3 n^3}$.

Hint: For a fixed 3-subset t of $[n]$ let Y_t be the Boolean r.v. which takes on the value 1 iff the triangle with vertex set t is in G . Note that $1 - Y_t$, viewed as a function $F_t(X_1, \dots, X_m)$ of the random edges, is coordinate-wise decreasing; see HW 6-Problem 4.

- (c) What lower bound do you obtain for $\mathbb{P}(X = 0)$ if you combine part (a) and (b)? How does this compare with the upper bound we obtained from Janson's inequality in class?

3. Let X be a random variable that takes values on a finite set S . Prove the following:

- (a) $0 \leq H(X) \leq \log_2 |S|$.

- (b) For any extraction function $\text{Ext} : S \rightarrow \{0, 1\}^*$ of X one has

$$\mathbb{E}[|\text{Ext}(X)|] \leq H(X),$$

where $|y|$ stands for the number of bits of a binary sequence y .

Hint: You may want to show that $\mathbb{P}(X = x) \leq 2^{-|\text{Ext}(x)|}$ for all $x \in S$.